HERIOT WATT UNIVERSITY



B37VR ROBOTICS PROJECT

2 Introduction to controlling motors using H Bridge

January '21

2 SUMMARY

- **LEARNING RESOURCES AVAILABLE**
- **LEARNING ACTIVITIES**
  - Learning to control speed and direction of motors using H Bridge.
  - Show operation of motors
  - Build sensor circuit for the 2 light detectors
- **TASKS TO BE COMPLETED**
  - Set up motor to operate using adafruit-arduino-lesson-15-dc-motor-reversing lab sheet
  - Demonstrate adjusting speed of motor
  - Run motors in forward and reverse

H-BRIDGE CONTROLLING A MOTOR

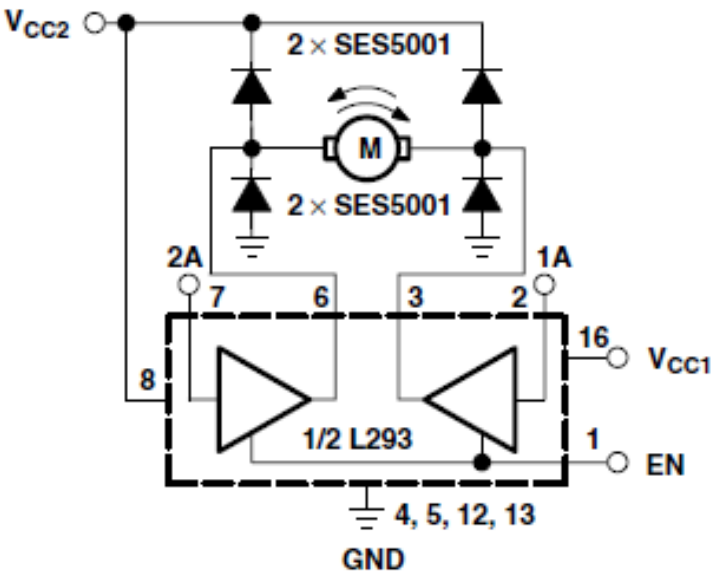
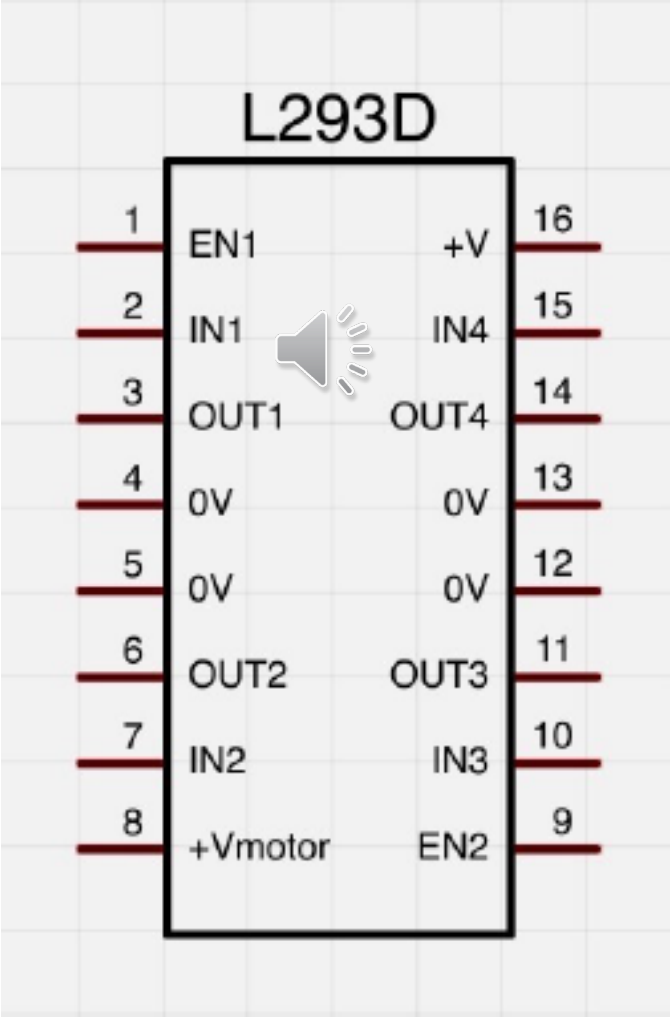


Figure 5. Bidirectional DC Motor Control

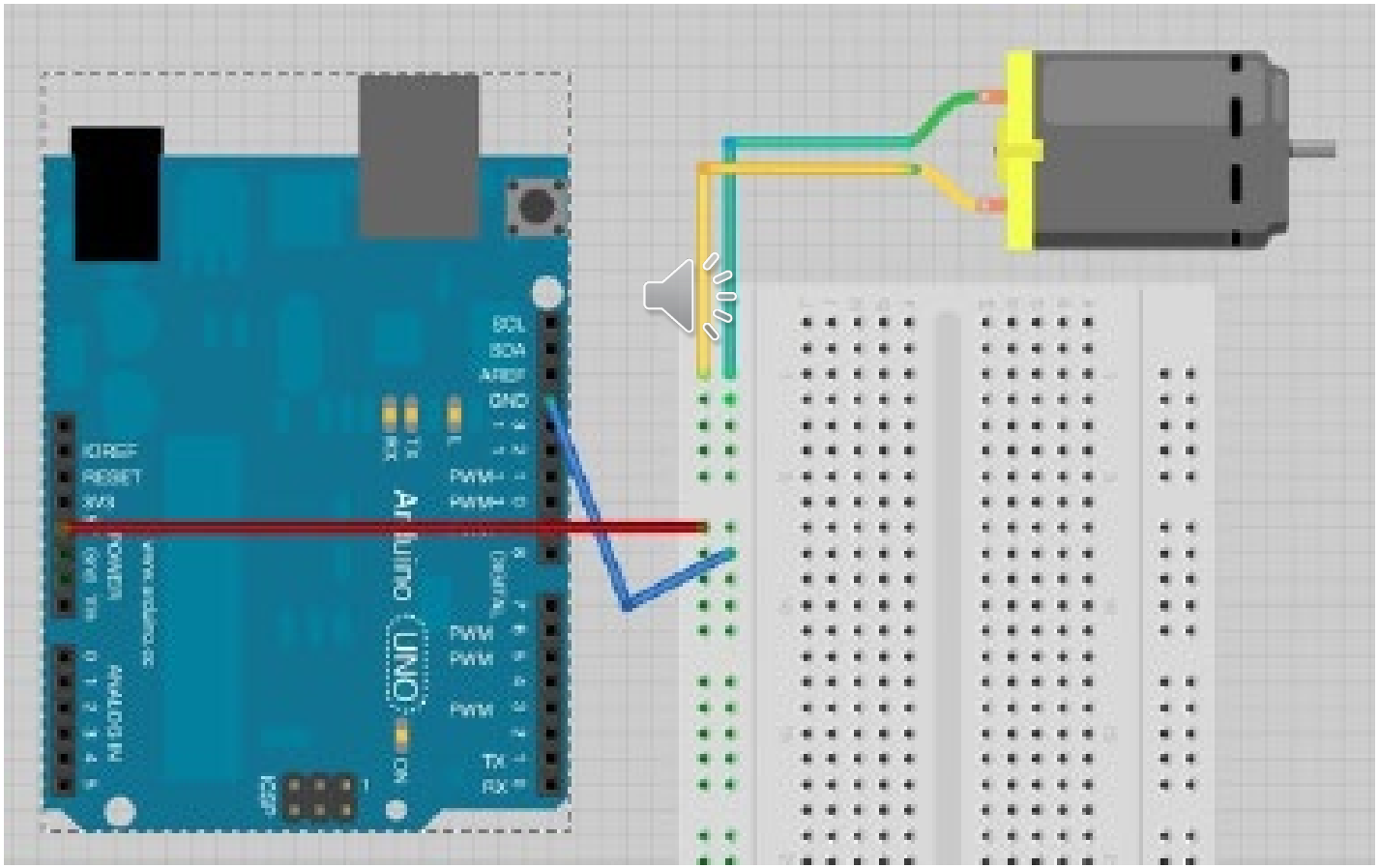
EN	1A	2A	FUNCTION
H	L	H	Turn right
H	H	L	Turn left
H	L	L	Fast motor stop
H	H	H	Fast motor stop
L	X	X	Fast motor stop

L = low, H = high, X = don't care

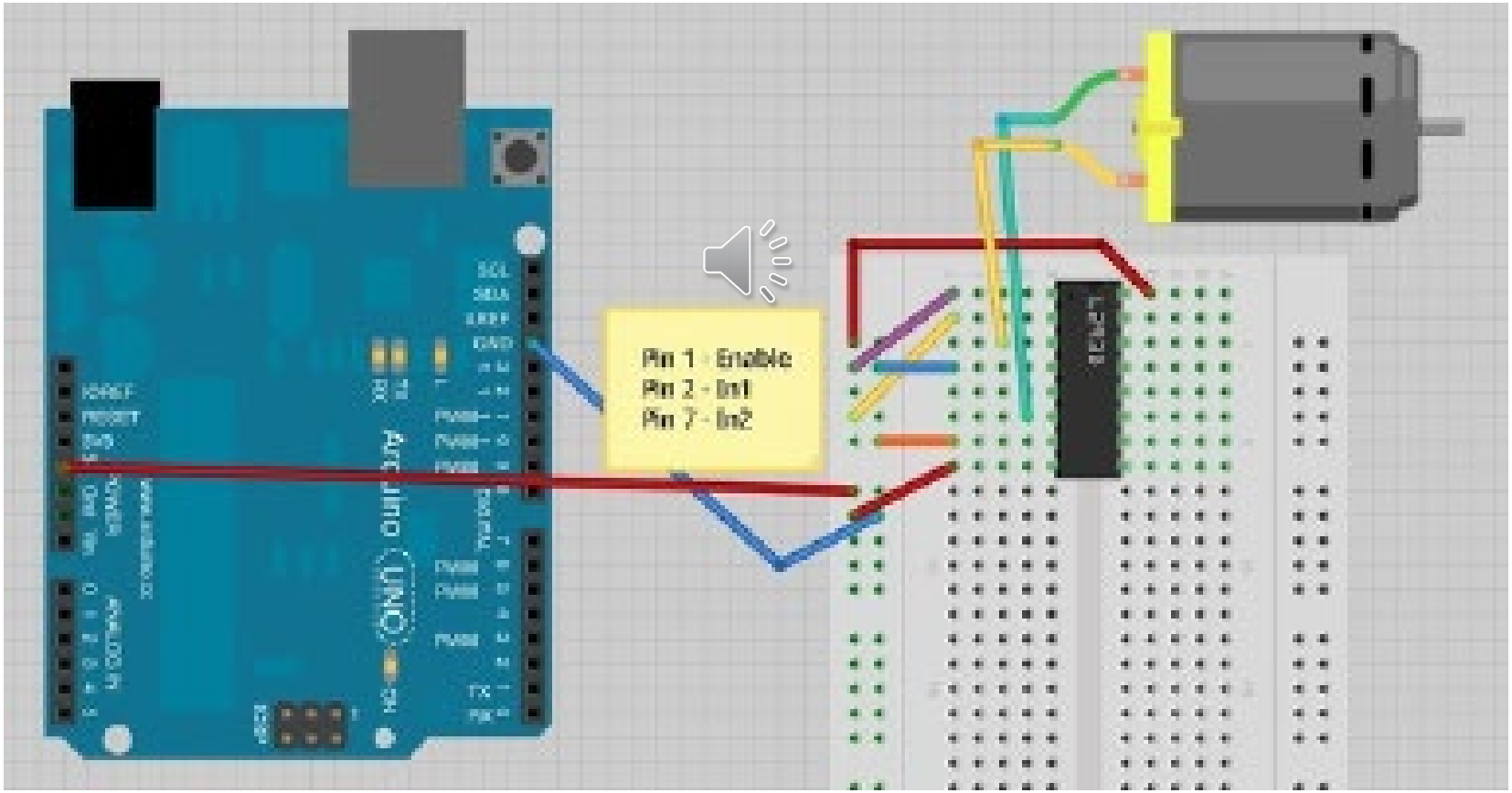
L293 H-BRIDGE



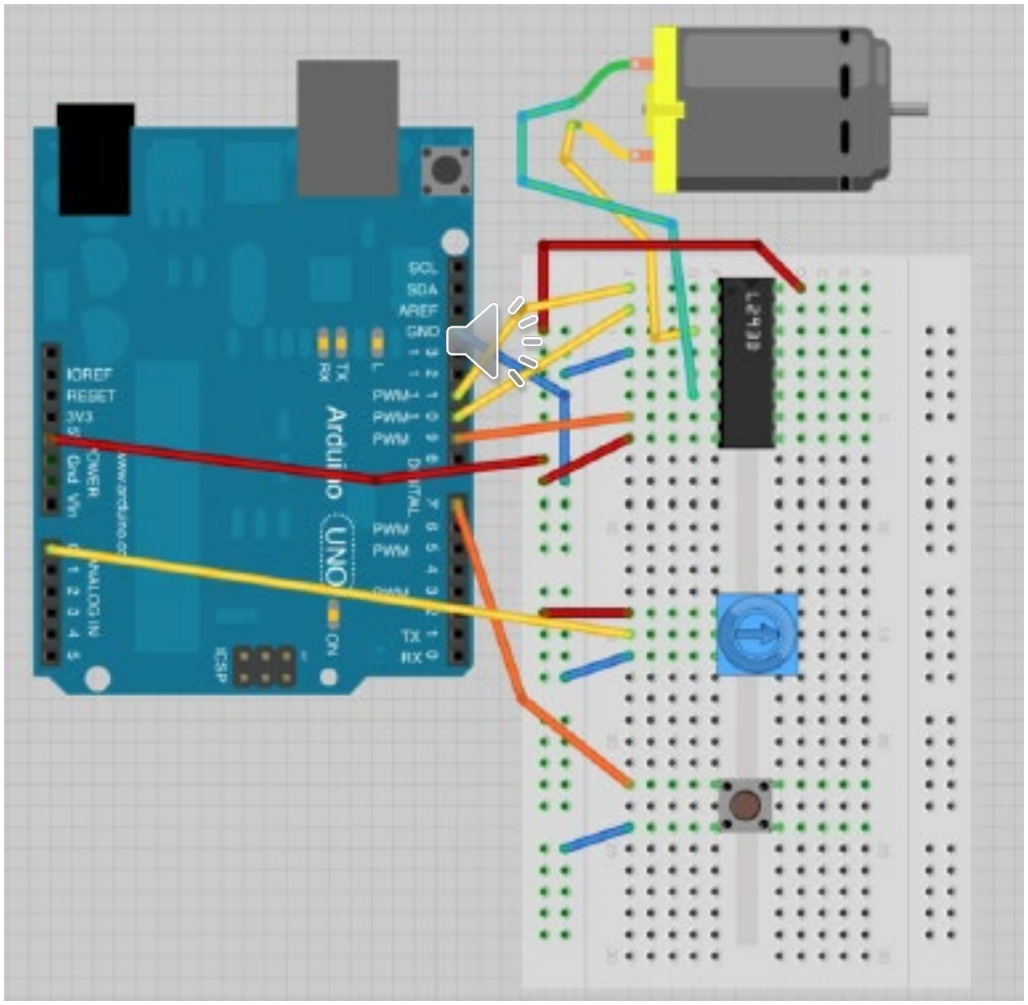
RUNNING A DC MOTOR DIRECTLY FROM A POWER SUPPLY



RUNNING A DC MOTOR FROM AN H-BRIDGE BUT NOT CONTROLLONG SPEED



RUNNING A DC MOTOR CONTROLLING SPEED AND DIRECTION



CODE

```
/*
Adafruit Arduino - Lesson 15. Bi-directional Motor
*/

int enablePin = 11;
int in1Pin = 10;
int in2Pin = 9;
int switchPin = 7;
int potPin = 0;

void setup()
{
  pinMode(in1Pin, OUTPUT);
  pinMode(in2Pin, OUTPUT);
  pinMode(enablePin, OUTPUT);
  pinMode(switchPin, INPUT_PULLUP);
}

void loop()
{
  int speed = analogRead(potPin) / 4;
  boolean reverse = digitalRead(switchPin);
  setMotor(speed, reverse);
}

void setMotor(int speed, boolean reverse)
{
  analogWrite(enablePin, speed);
  digitalWrite(in1Pin, ! reverse);
  digitalWrite(in2Pin, reverse);
}
```

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